How does BERT capture semantics? A closer look at polysemous words

David Yenicelik
ETH Zürich
yedavid@ethz.ch

Florian Schmidt
ETH Zürich
florian.schmidt@inf.ethz.ch

Yannic Kilcher
ETH Zürich
yannic.kilcher@inf.ethz.ch

Abstract

The recent paradigm shift to contextual word embeddings has seen tremendous success across a wide range of down-stream tasks. However, little is known on how the emergent relation of context and semantics manifests geometrically. We investigate polysemous words as one particularly prominent instance of semantic organization. Our rigorous quantitative analysis of linear separability and cluster organization in embedding vectors produced by BERT shows that semantics do not surface as isolated clusters but form seamless structures, tightly coupled with sentiment and syntax.

1 Introduction

Word embeddings have not only proven to be excellent representations in standalone tasks (Mikolov et al., 2013; Pennington et al., 2014; Wang et al., 2019) but have revolutionized the way modern NLP architectures are built (Collobert et al., 2011), and by now encode text input for virtually every task available. Recently, this approach has been paired with the transformer architecture (Vaswani et al., 2017) and a selection of pre-training tasks to bootstrap more powerful *contextual* word embeddings such as the ones produced by BERT (Devlin et al., 2018). s

The paradigm of encoding a word in its context has elevated the embedding methodology once more and from several perspectives. First, performance improvements on down-stream tasks are extraordinary across a wide range of tasks (Ethayarajh, 2019; Devlin et al., 2018; Wang et al., 2018). Second, the embedding space now must incorporate a vastly larger number of vectors, and its organization becomes an interesting research question on its own, especially given the largely unattributed performance gains.

In this work, we investigate the important concept of *polysemy* as one prominent example of semantic sub-space organization. Given that a word such as 'bank' can have several meanings, how are the corresponding vectors arranged in a contextual embedding space?

We investigate the organization of polysemous words in BERT embeddings through the concepts of separability and clusterability using the WordNet annotations in SemCor (Miller et al., 1990). Our particular focus is a rigorous quantitative rather than purely qualitative analysis.

2 Related Work

Work connecting polysemy and word vector representations is often limited to static word embeddings where context has to be re-introduced through graph-based approaches (Remus and Biemann, 2018), auxiliary corpora (Pelevina et al., 2016), or even image data (Bruni et al., 2013). Usually, word sense disambiguation (WSD) performance is then chosen as a proxy to semantic disambiguation (Pilehvar and Camacho-Collados, 2019), yet no insights into the organization of the vector space are obtained. In a similar spirit, Kageback and Salomonsson (2016) add context through a recurrent encoder, yet do not analyze the geometry of these encodings.

In the meantime, the WSD task has been tackled successfully with BERT embeddings and Wiedemann et al. (2019) show that even a non-parametric approach suffices, which confirms that BERT must arrange word vectors according to semantic properties and suggests that no additional semantic pre-training is necessary (Levine et al., 2019).

When BERT embeddings of a polysemous word are analyzed, the findings are often summarized as a Silhouette score (Rousseeuw, 1987) or custom

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variance measures (Ethayarajh, 2019). While this allows to compare the average displacement due to semantic change across words, it does not give us a good sense of the overall structure of word vectors. In addition, the embedding space produced by BERT has been analyzed in terms of syntactic features, such as parse-trees (Coenen et al., 2019; Jawahar et al., 2019), part-of-speech, verbs and arguments (Shi et al., 2019; Ribeiro et al., 2019).

It is clear that BERT distinguishes polysemous words at least locally by nearest neighbors (Schmidt and Hofmann, 2020). However, the extent to which clusters are formed and how they are connected has only been addressed qualitatively (Coenen et al., 2019; Jawahar et al., 2019; Wiedemann et al., 2019), and no agreed-upon answer has emerged. This can be partly attributed to their qualitative methodology.

3 Method

How can we verify a hypothesis about the organization of polysemous words without manually inspecting the geometry for each word? Given a set of sentences with annotated polysemy, two strategies emerge: First, we can inspect the embedding space through the lens of a classifier with a clearly defined hypothesis set and take its accuracy as a signifier for the corresponding organization. Second, we can use an unsupervised approach to detect sub-space organization and compare its result to the WordNet labels using an appropriate similarity metric. We will proceed by analyzing both questions.

In our experiments, we consider the output of the last layer of BERT as the contextual word embeddings since this layer is most commonly used for downstream tasks, as depicted in Figure 1. To work with a discrete formalization of semantics, we use the WordNet 3.0 annotations in the SemCor 3.0 sentence dataset (Miller et al., 1990). This allows us to retrieve embeddings that are annotated with a ground-truth *semantic class* label. SemCor is one of the largest sense-annotated corpora with $37,176$ sentences, enabling us to quantify semantics and sample labelled word embeddings.

3.1 Linear Separability

Before turning to the clustering task, we investigate to what degree semantic classes can be separated by a hyperplane in embedding space, resulting in *semantic regions* as depicted in Figure 2. To this end, we train a simple linear classifier on top of the BERT embeddings (without fine-tuning) to predict the semantic class and report accuracy. Crucially, we down-project the 768-dimensional vectors using PCA ensuring that separability is not merely a consequence of high dimensionality. In the high-dimensional setting, this allows to assess to what extent semantic regions do form.

3.2 Clusterability

Once the presence of semantic regions has been concluded, we want to investigate the extent to which clusters form and how they are connected. For this, we train clustering models to understand the modality of the data, and to what extent clusters are in isolation from each other.

Because we are interested in practical gains, we refrain from using purely theoretical tools and clusterability scores (Ackerman and Ben-David, 2009; Mccarthy et al., 2016). In contrast, we use interpretable clustering models (Frey and Dueck, 2007; Ester et al., 1996; Campello et al., 2013; Comani-

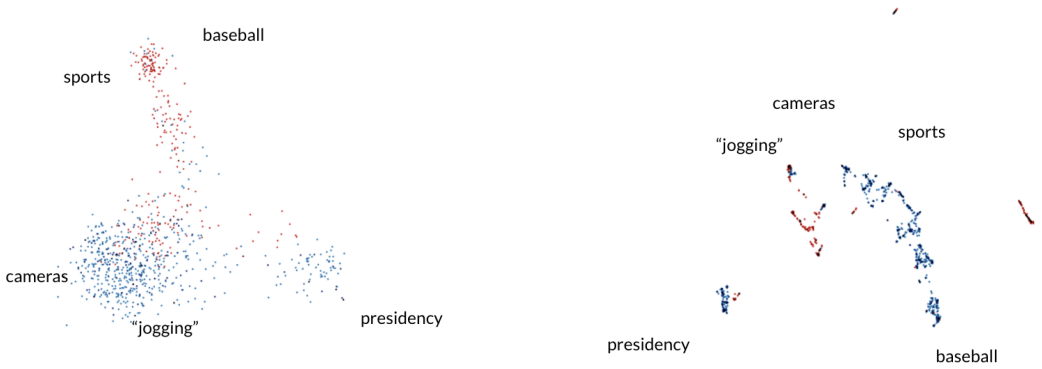

Figure 1: PCA (left) and UMAP (right) visualizations for contextual word embeddings sampled for the word `run`. Red points denote *nouns*, blue points denote *verbs*

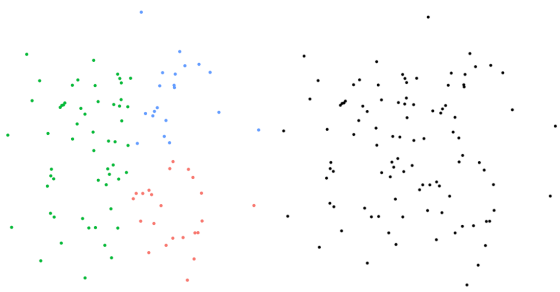

Figure 2: Datapoints generated by sampling from a normal distribution. (left) includes class information denoted by the green, blue and red colors, forming semantic regions. This data is linearly separable, as there is always a hyperplane separating any two classes. (right) The class label information is not available. It is unclear how a clustering would look like.

ciu and Meer, 2002) that can detect the number of clusters in the data, as well as an adapted version of the Chinese Whispers algorithm (Biemann, 2006) that accounts for the *hubness property* [1] amongst embedding vectors outputted by BERT. The Chinese Whispers algorithm relies on a graph produced by the word embeddings and identifies clusters by passing messages between the nodes of the graph. From the sampled word embeddings we create the graph adjacency matrix M by calculating the pairwise cosine similarity between embeddings, and similar to Ribeiro et al. (2019), prune any edges which correspond to a cosine similarity lower than $w_{\text{cutoff}} = \mu(M) + c\sigma(M)$ where c is a hyperparameter, and μ and σ are the mean and standard deviation of all cosine similarities recorded in M. Hubs are defined as the top n embedding vectors with highest cumulative cosine similarities.
The development and test sets consist of $\frac{n}{2}$ words respectively, including their set of sampled embedding vectors.

To score the overlap between a predicted clustering and the underlying ground-truth labels, we use the Adjusted Random Index (ARI) (Rand, 1971; Hubert and Arabie, 1985), which returns a similarity measure where a value of 1 implies an identical clustering up to a permutation and a value of 0 implies random predictions. Please note that this also introduces a small penalty when more clusters are introduced than actually present in the dataset according to the cluster-class-labels. However, preventing this is not in the scope of this work, and as such we do not further investigate this.

If no clustering is found, one can say with high confidence that the semantic regions are not occurring in different modes, and rather transition seamlessly into one another. This allows for an assessment in high dimensional space to what extent semantic regions are obvious, apparent by distinct modes. The motivation behind both experiments is visually depicted in Figure 2.

4 Experiments

We proceed with a discussion of the experimental results.

4.1 Bias in SemCor

First we aim to develop an understanding of bias in SemCor, as any bias in the data will propagate on to further observations. We conduct a simple experiment where we analyse the distribution of occurrences of semantic class ids.

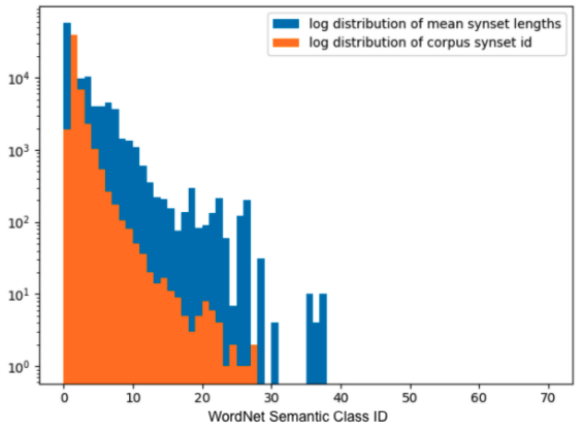

Figure 3: A cumulative plot over all words with WordNet senses within SemCor 3.0 and their respective frequencies. The SemCor data is biased. Words with a low WordNet sense index, i.e. close to 0, occur more often than words with a high WordNet sense index, i.e. above 5. There would be no bias if the two distributions would overlap. The skew could be a natural effect of how words with lower WordNet indices are assigned to more frequently used words.

Figure 3 depicts that the SemCor corpus is biased towards semantic classes which have a lower WordNet class ID. This could be due to the nature of WordNet, likely assigning low id indices to frequently used words. This requires us to oversample underrepresented classes for select experiments.

[1] *hubs* are embeddings close to a majority of other embedding vectors, degrading performance (Conneau et al., 2017)